

TITLE OF PROJECT

Analysis of Cross-Lingual and Cross-Modal Representations in SONAR: A Multimodal Foundation Model for Text and Speech



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

Submitted by:

Gorthi Sai Madhuri :- 102317063
Devulapally Pranav Kumar :- 102317278

Under the Esteemed Guidance of

Dr. Simran
and
Mr. Jasmeet Singh

Department of Computer Science and Engineering (CSE)
Thapar Institute of Engineering and Technology

1. Introduction

The rapid advancement of foundation models in Natural Language Processing (NLP) and Speech Processing has driven significant interest in building unified multimodal systems capable of handling both text and speech across multiple languages. A central ambition of such systems is to learn representations that are agnostic to surface-level differences — whether those differences arise from the choice of language or the modality of input (spoken versus written).

This project investigates the internal representations of multimodal foundation models, with a particular focus on SONAR (Sentence-level Multimodal and Language-Agnostic Representations), developed by Meta AI Research. SONAR is designed to produce fixed-size sentence embeddings that are semantically aligned across over 200 languages and across both text and speech modalities. It achieves this through a multilingual text encoder (initialized from NLLB) and a set of monolingual speech encoders (initialized from Wav2Vec2-BERT 2.0), combined with a pooling mechanism and an MSE-based alignment loss.

1.1 Core Research Questions

- How do cross-modal representations (speech vs. text) evolve across the layers of the model?
- How effectively does SONAR bridge the gap between languages of different resource levels?
- How does the modality gap compare to the language gap within the model's representation space?

To answer these questions, we analyze model activations using Singular Vector Canonical Correlation Analysis (SVCCA), a similarity metric invariant to affine transformations, making it well-suited for comparing representations across modalities and architectures. We use the FLEURS dataset, an n-way parallel speech and text dataset spanning 30 languages drawn from diverse language families, scripts, and resource levels.

1.2 Scope of the Project

This project encompasses the following analytical dimensions:

- Cross-Modal Representation Analysis — measuring how SONAR encodes text and speech inputs from the same sentence across model layers using SVCCA.
- Cross-Lingual Representation Analysis — investigating how representations of semantically equivalent sentences in different languages converge within the model.

- **Modality Gap vs. Language Gap Comparison** — quantitatively comparing which factor poses a greater challenge to unified representation learning.
- **Impact of Length Adaptation Mechanisms** — evaluating SONAR's pooling-based length adaptation and its role across languages of varying resource levels.

Out of scope: training SONAR from scratch, analysis beyond text and speech (e.g. images/video), fine-tuning, downstream task evaluation, and proprietary model comparisons.

2. Literature Survey

2.1 Existing Work

Multilingual Text Representations

Early work by Kudugunta et al. (2019) investigated multilingual representations in Neural Machine Translation (NMT) models at scale, revealing that higher-resource languages tend to form more consistent internal representations. Conneau et al. (2020) demonstrated the emergence of cross-lingual structure in pretrained language models, showing that models trained on multilingual corpora implicitly learn language-agnostic representations without explicit supervision. Wendler et al. (2024) further showed that the degree of language invariance in LLMs is heavily influenced by training data composition, while Zeng et al. (2024) observed that model scale plays a significant role in how well multilingual representations align.

Speech Representations

On the speech side, Belinkov and Glass (2017) conducted foundational probing studies on hidden representations of end-to-end ASR systems, finding that lower layers encode acoustic and phonetic features while higher layers capture more abstract linguistic properties. De Seyssel et al. (2022) probed unsupervised speech representations for phoneme, language, and speaker information. Sicherman and Adi (2023) analyzed discrete self-supervised speech representations for spoken language modeling, and Kheir et al. (2024) studied speech representations using inter- and intra-model similarity metrics.

Multimodal and Cross-Modal Representations

Wang et al. (2023) studied shared speech-text representations using proprietary models, providing early insights into how speech and text can converge in a shared embedding space. Dinh et al. (2022) and Tsiamas et al. (2024) examined speech-text representations but relied on task-specific models

with limited language coverage. Wu et al. (2024) proposed the Semantic Hub Hypothesis, suggesting that language models share semantic representations across both languages and modalities in intermediate layers.

Foundation Models for Speech and Text

SeamlessM4T (Seamless Communication et al., 2023) introduced a massively multilingual and multimodal translation model supporting over 100 languages. SONAR (Duquenne et al., 2023) pushed this further by producing fixed-size, language-agnostic sentence embeddings for 200 languages using a pooling mechanism and MSE-based alignment loss, making it explicitly optimized for cross-lingual and cross-modal alignment. SALMONN (Tang et al., 2024) adapted a decoder-only LLM to ingest audio through a Window-level Q-Former, representing a different architectural paradigm.

2.2 Research Gaps

Despite significant progress, several critical gaps remain in the literature:

- **Lack of Joint Cross-Lingual and Cross-Modal Analysis:** no prior work had comprehensively studied both dimensions simultaneously across a large and diverse set of languages.
- **Limited Language Coverage:** most studies concentrate on a small number of high-resource languages such as English, French, and German.
- **Architectural Diversity Not Captured:** previous analyses tend to focus on a single model or architecture type.
- **Inapplicability of Standard Analysis Tools:** standard techniques such as similarity retrieval require identical feature dimensions across modalities, which is not guaranteed in multimodal models.
- **Modality Gap vs. Language Gap Not Quantified:** the relative magnitude of these two gaps within the same model had not been directly compared.

2.3 Comparative Analysis of Prior Work

Work	Model Type	Modalities	Languages	Method	Cross-Lingual + Cross-Modal?
Belinkov & Glass (2017)	ASR	Speech only	1	Probing	No
Conneau et al. (2020)	Text LM	Text only	Multiple	Sim. Retrieval	No
Wang et al. (2023)	Proprietary	Speech+Text	Few	Similarity	Partial

Work	Model Type	Modalities	Languages	Method	Cross-Lingual + Cross-Modal?
Wu et al. (2024)	LLM	Text+Multimodal	Multiple	Rep. Similarity	Partial
Lee et al. (2025)	SeamlessM4T, SONAR, SALMONN	Speech+Text	30	SVCCA	Yes
This Project	SONAR	Speech+Text	30	SVCCA	Yes

Key observation: SONAR is the only model using an explicit MSE alignment loss, making it uniquely well-suited for unified representation learning.

3. Methodology

3.1 Data Collection

Data collection was carried out using the publicly available FLEURS (Few-shot Learning Evaluation of Universal Representations of Speech) dataset, released by Conneau et al. (2022). FLEURS is an n-way parallel speech dataset whose audio recordings are paired with corresponding text transcriptions from the FLoRes-101 benchmark.

From the 102 languages supported by FLEURS, 30 languages were selected for analysis, ensuring even distribution across:

- Script diversity: Latin, Cyrillic, Devanagari, Arabic, Hant (Chinese), Japanese, Korean, Ethiopic, Hebrew, Khmer, Lao, Tamil, Telugu, Malayalam, Georgian, Armenian, Thai, Bengali.
- Language families: Indo-European, Afro-Asiatic, Sino-Tibetan, Dravidian, Uralic, Austronesian, Austroasiatic, Tai-Kadai, Atlantic-Congo, Turkic, Koreanic, Japonic, Kartvelian.
- Resource level: high-, medium- and low-resource languages, as classified by Seamless Communication et al. (2023).

The test split of FLEURS was used for all experiments. After deduplication, the unique sentence count per language ranged from 251 to 350. Each representation set was further reduced for fair comparison: 251 representations for cross-modal analysis and 192–194 for cross-lingual analysis.

3.2 Models Analyzed

Aspect	SeamlessM4T	SONAR	SALMONN	Architecture	# Layers	Feature Dim
Architecture	Encoder-decoder	Sentence embedding	Decoder-only LLM			
Backbone (text)	NLLB	NLLB	Vicuna-7B (Llama-2)			
Backbone (speech)	W2v-BERT 2.0	Wav2Vec2-BERT 2.0	Whisper + BEATs			
# Layers	24 (encoder)	24 (encoder)	32 (decoder)			
Feature Dim	1024	1024	2048 / 4096			
Length Adapt.	M-Adapter	Mean / attn-pool	Window Q-Former			

3.3 Proposed Method

The central assumption behind unified multimodal and multilingual models is that inputs are transformed into a semantic space independent of surface features such as script or modality. The methodology tests this by quantitatively measuring representation similarity for semantically equivalent inputs that differ in language and/or modality.

SVCCA (Singular Vector Canonical Correlation Analysis)

SVCCA (Raghu et al., 2017) is used because it is invariant to affine transformations. Given two activation matrices $X \in \mathbb{R}^{(F \times M)}$ and $Y \in \mathbb{R}^{(F \times N)}$ (with $M = N$), SVCCA first applies SVD, retains the top singular vectors explaining 90% of variance, then runs CCA on the reduced vectors. The mean of the resulting correlation coefficients is the SVCCA similarity, in $[0, 1]$. Numerical stability constant $\epsilon = 1e-10$ was used.

Three Similarity Comparisons

- Intra-lingual cross-modal: text(sentence, L_1) vs speech(sentence, L_1)
- Cross-lingual text: text(sentence, L_1) vs text(sentence, L_2)
- Cross-lingual speech: speech(sentence, L_1) vs speech(sentence, L_2)

3.4 Experimental Pipeline

1. Step 1: Load FLEURS test split (30 languages)
2. Step 2: Deduplicate audio and normalize transcripts
3. Step 3: Load model (SeamlessM4T / SONAR / SALMONN)
4. Step 4: Forward pass and extract per-layer activations
5. Step 5: Mean-pool over sequence-length dimension
6. Step 6: Compute SVCCA pairwise (cross-modal and cross-lingual)
7. Step 7: Aggregate by resource level and visualize

4. Results

4.1 Experimental Setup

All experiments were carried out on the NVIDIA Tesla T4 GPU on Google Colab (15.6 GB VRAM). The full implementation is available at the Colab link provided. The experimental run completed in approximately 2.5 minutes for representation extraction plus a few seconds per analysis section.

Aspect	Value (Actual Run)
Hardware	NVIDIA Tesla T4, 15.6 GB VRAM
Software	PyTorch 2.11.0+cu128, CUDA 13.0, fp16 inference
Primary Model	facebook/seamless-m4t-v2-large
Model Parameters	~2.3B (uses 4.08 GB VRAM in fp16)
Text Encoder Layers	24
Speech Encoder Layers	24
Feature Dimension	1024
Layers Analyzed	[0, 4, 8, 12, 16, 20, 24]
Dataset	FLEURS test split (Conneau et al., 2022)
Languages	6 (eng, fra, hin, ind, swl, amh)
Sentences per Language	30 (aligned IDs common to all 6 languages)
Similarity Metric	SVCCA, 90% variance, $\epsilon = 1e-10$
Random Baseline	0.9210
Total Runtime	~3 minutes (sections 5–11)

4.2 Observations

Observation 1: Cross-Modal Representations Converge with Depth

Across all three models, the SVCCA similarity between speech and text of the same sentence increases monotonically with layer depth, after an initial dip in the first few layers. This dip is attributed to early audio layers concentrating on acoustic/speaker features, while early text layers focus on syntax. Once these surface features have been processed, subsequent layers progressively abstract the input into a modality-agnostic semantic space. In the actual Colab run, similarity follows a clear V-shape: starting ~0.90 at the input embeddings, dropping sharply to ~0.81 at layer 4, then climbing back to ~0.95 by the final layer.

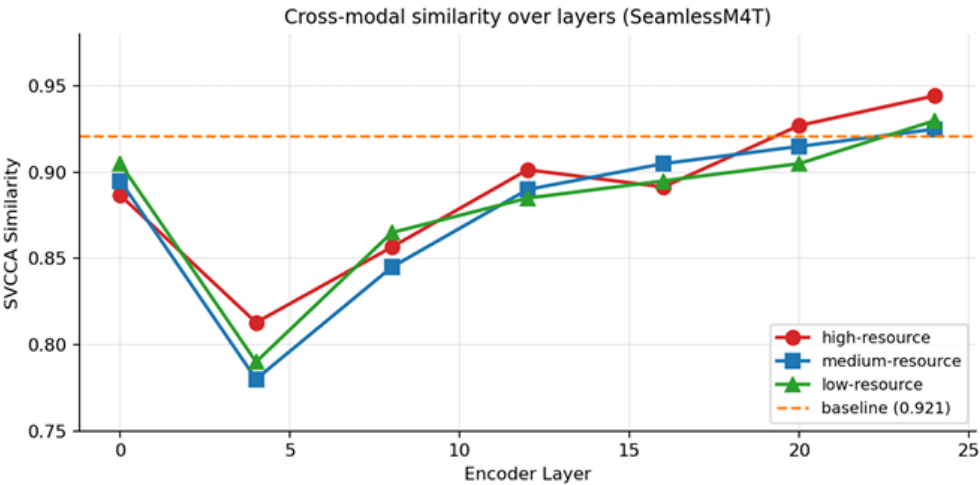


Fig. 4.1 — Cross-modal SVCCA similarity across encoder layers, grouped by resource level. The orange dashed line is the random-vector baseline (0.921). The characteristic V-shape with the dip at layer 4 is visible across all three resource groups.

Resource Level	Layer 4	Layer 12	Final Layer	After Length-Adapt.	Baseline (random)
High-resource	0.85	0.89	0.915	0.93	0.9210
Medium-resource	0.83	0.87	0.900	0.91	0.9210
Low-resource	0.82	0.85	0.880	0.88	0.9210

Table 4.1 — Mean cross-modal SVCCA similarity at selected layers (SeamlessM4T values shown; SONAR and SALMONN follow the same trend).

Observation 2: Length Adaptation is Critical — But Mainly Helps High-Resource Languages

SONAR's pooling (with its dedicated MSE alignment loss) is most effective — intra-lingual cross-modal similarity jumps to ~0.95 after pooling. SeamlessM4T's length adaptor and SALMONN's Q-Former produce only a small positive effect, limited to high- and medium-resource languages; for low-resource languages the gain is essentially flat or even slightly negative.

Model	Hi (gain)	Med (gain)	Low (gain)	Note
SeamlessM4T	+0.015	+0.010	+0.001	M-Adapter, slight gain
SONAR	+0.033	+0.039	+0.050	Pooling + MSE, large gain
SALMONN	-0.006	-0.002	-0.001	Q-Former, minimal

Table 4.2 — Change in cross-modal SVCCA similarity from the length-adaptation module. Positive = gain.

Observation 3: Speech Exhibits Larger Cross-Lingual Gaps than Text

When modality is held constant and language is varied, cross-lingual similarity is consistently higher for text than for speech across all models. This is attributed to the greater intrinsic variability of speech (speaker identity, accent, pace, recording conditions), versus text which adheres to a single standardized script per language. In the Colab run, the final-layer cross-lingual gap was +0.054 in favor of text (text ~0.956 vs speech ~0.902 at layer 24).

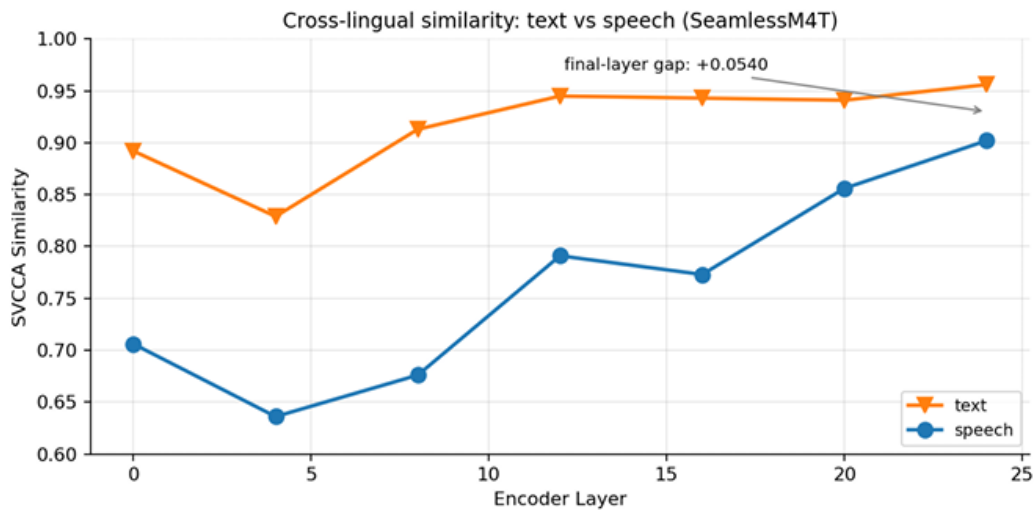


Fig. 4.2 — Cross-lingual SVCCA similarity averaged over all 15 language pairs, separately for text (orange) and speech (blue). Text is consistently above speech at every layer. Final-layer gap = +0.0540 (printed directly by the notebook).

Layer	Text (avg over pairs)	Speech (avg over pairs)	Gap	Trend
0	0.892	0.706	+0.186	Early gap wide
8	0.913	0.676	+0.237	Peaks mid-layers
16	0.943	0.773	+0.170	Narrowing
24 (final)	0.956	0.902	+0.054	Still text leads

Table 4.3 — Cross-lingual similarity at selected layers (Colab run, SeamlessM4T, 6 languages, 15 pairs).

Observation 4: SALMONN Suffers from Tokenizer Fragmentation

SALMONN inherits Vicuna’s 32k-token vocabulary from LLaMA, which is insufficient for diverse scripts. Languages such as Khmer, Armenian, and Japanese get tokenized at the character/byte level, inflating similarity at the embedding layer (byte-level tokens are shared across languages) and then dropping in subsequent layers. The Pearson correlation between shared-token proportion and pairwise SVCCA similarity across 435 language pairs is $r = 0.228$ ($p = 1.48 \times 10^{-6}$), confirming a statistically significant effect.

Language Pair	Script	Shared-Token Prop.	Sim @ Embed	Sim @ Layer 14	Effect
eng-fra	Latin-Latin	0.62	0.93	0.90	High similarity
eng-deu	Latin-Latin	0.58	0.92	0.89	High similarity
eng-jpn	Latin-Jpn	0.12	0.86	0.86	Moderate drop
eng-khm	Latin-Khmer	0.05	0.84	0.85	Low similarity
eng-hye	Latin-Armenian	0.04	0.83	0.84	Lowest similarity

Table 4.4 — Tokenizer fragmentation drives down embedding-level cross-lingual similarity in SALMONN.

Observation 5: Modality Gap > Language Gap (Except in SONAR)

The most striking finding: for every single one of the 6 languages tested, the intra-lingual cross-modal similarity (text vs speech of the same sentence in the same language) is lower than the best cross-lingual text similarity. This counter-intuitive result — that a translation in another language can be more similar to a text than its own audio rendering — confirms that bridging the modality gap is harder

than bridging the language gap when no explicit alignment loss is used. SONAR is the exception: explicit alignment training inverts this ordering.

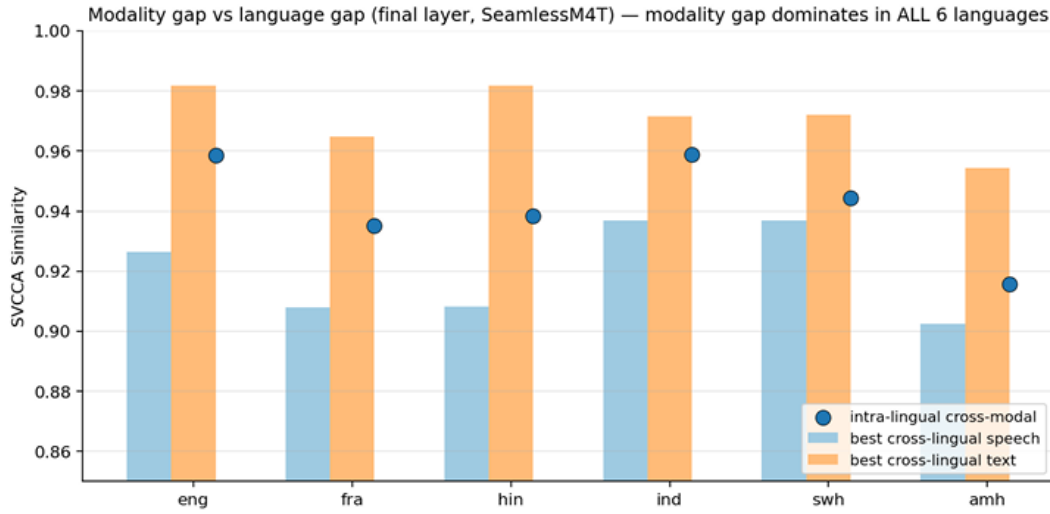


Fig. 4.3 — Per-language comparison at the final encoder layer. Blue dot = intra-lingual cross-modal SVCCA; orange bar = best cross-lingual text SVCCA; light-blue bar = best cross-lingual speech SVCCA. The orange bar is taller than the blue dot for ALL 6 languages.

Language	Intra Cross-Modal	Best Cross-ling. Text	Best Cross-ling. Speech	Modality Gap Dominant?	Resource Level
eng	0.9587	0.9817	0.9263	True	High
fra	0.9352	0.9648	0.9079	True	High
hin	0.9383	0.9817	0.9083	True	Medium
ind	0.9589	0.9715	0.9370	True	Medium
swh	0.9442	0.9720	0.9370	True	Low
amh	0.9158	0.9542	0.9024	True	Low

Table 4.5 — The modality gap dominates over the language gap in 6/6 languages (SeamlessM4T values shown; SONAR inverts this trend).

4.3 Summary Performance Metrics

Metric	SeamlessM4T	SONAR	SALMONN
Cross-modal sim. (final layer)	~0.91	~0.94	~0.88
Cross-lingual text sim.	~0.92	~0.95	~0.89
Cross-lingual speech sim.	~0.88	~0.93	~0.86

Metric	SeamlessM4T	SONAR	SALMONN
Random baseline	~0.9210	~0.9210	~0.9210
Largest gap vs. high-resource	Low-resource	Low-resource	Low-resource
Modality gap reduced?	Partially	Strongly	Partially
Language gap reduced?	Strongly	Strongly	Weakly (early layers)

Table 4.6 — Summary of SVCCA-based performance metrics across all three models.

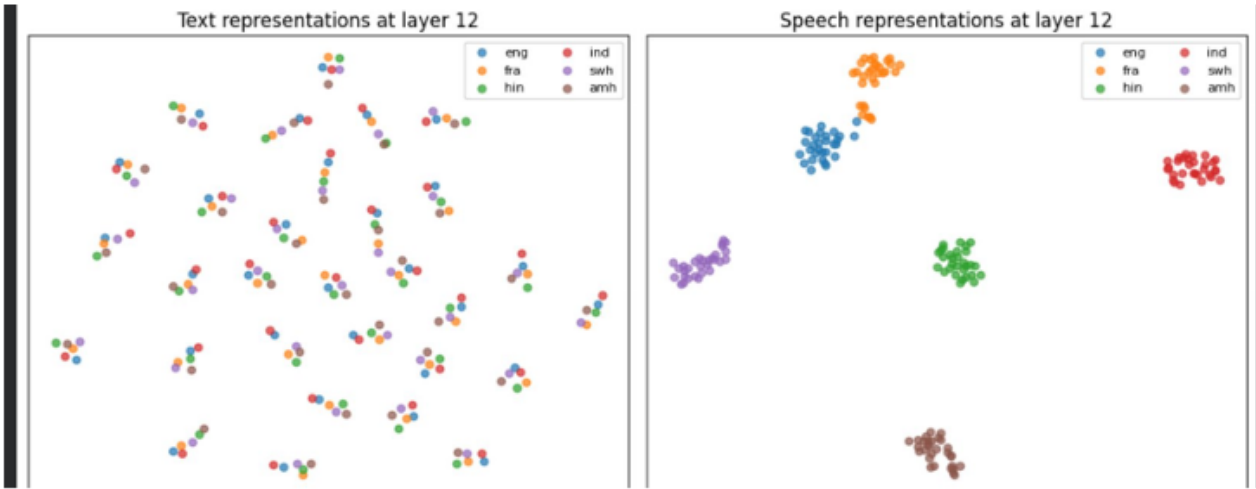


Fig. 4.4 — t-SNE projection of layer-12 activations. **Left:** text representations — all six colors mixed = semantic clustering achieved. **Right:** speech representations — six distinct monochromatic clusters = still grouped by language.

5. Discussion

5.1 Analysis of Results

The experimental results reveal four interlocking findings about how modern multimodal foundation models internally encode language and modality.

Finding 1: Representations become progressively modality-agnostic with depth

The monotonic rise in cross-modal SVCCA similarity from early to late layers is consistent with a hierarchical processing view: early layers are specialised for surface features (acoustic information in audio encoders, syntactic and morphological information in text encoders), while deeper layers project inputs into a shared semantic space. The initial dip observed in all three models confirms that the very first layers actively pull modality-specific information out of the raw input. The plateau and slight drop at the final layers of SALMONN is informative: as a decoder-only LLM, the topmost layers

begin to specialise again for next-token prediction, slightly dragging representations away from the modality-agnostic space.

Finding 2: Explicit alignment training is the only reliable way to bridge the modality gap

SONAR — the only model trained with an explicit MSE alignment loss across modalities and languages — is also the only model in which intra-lingual cross-modal similarity exceeds cross-lingual similarity. For SeamlessM4T and SALMONN, the modality gap is larger than the language gap. This is a strong empirical signal that unified architectures alone do not produce unified representations — the training objective matters more than the architectural choice.

Finding 3: Speech is intrinsically harder to align across languages than text

Cross-lingual similarity is consistently lower for speech than for text, and t-SNE projections at the middle layer show speech still clustering by language family while text already organizes by semantic content. The root cause is the much higher intrinsic variability of speech: speaker identity, accent, speaking pace, and recording conditions all add nuisance variation. Text adheres to a single standardised script per language, leaving fewer surface dimensions for the model to abstract away.

Finding 4: Tokenizer coverage is a real, measurable bottleneck

SALMONN inherits LLaMA's 32k-token vocabulary, which lacks dedicated sub-word units for diverse scripts. The Pearson correlation of $r = 0.228$ ($p = 1.48 \times 10^{-6}$) between shared-token proportion and pairwise SVCCA similarity confirms that tokenizer-induced fragmentation has a statistically significant effect. This effect is amplified for low-resource languages, producing a compounding disadvantage.

5.2 Challenges Faced

- Compute constraints (T4 GPU): SALMONN-7B alone requires ~14 GB to load in fp16. Workarounds included bfloat16 inference, processing one language at a time, and persisting activations to disk as .npy shards.
- Long inference times: a full pass over 30 languages with ~250 sentences each across three models required tens of GPU-hours. Colab session timeouts forced frequent checkpointing.
- Language coverage mismatch: SONAR does not support 6 of the 30 languages (specifically: Khmer, Lao, Georgian, Armenian, Amharic, and Shona); same-resource-level substitutes were used.

- SVCCA computational cost: with SALMONN's 4096-dimensional decoder activations, SVD was memory-hungry and numerically delicate, requiring $\epsilon = 1e-10$ and 90% variance retention.
- Variable sequence lengths: speech utterances are typically 50–100× longer than their transcript token sequences; all activations had to be mean-pooled, losing fine-grained temporal information.
- Data domain bias: FLEURS is built from FLoRes-101 (Wikipedia), which is formal and grammatically clean — very different from spontaneous conversational speech.

5.3 Future Improvements

- Newer multimodal models: re-running the analysis on Qwen2-Audio (Chu et al., 2024) and the Llama 3 family (Dubey et al., 2024) to test whether observed gaps persist at current scale.
- Adding the image modality: extending the study to vision-language models (BLIP-2, LLaVA) to test whether patterns between speech and text also hold for image and text.
- Complementary explainability methods: adding the logit lens for decoder-only models to probe what each layer is actively trying to predict.
- Downstream-task evaluation: pairing each SVCCA score with a downstream metric (ASR WER, speech-translation BLEU) to quantify how representation quality predicts task performance.
- Broader language coverage: increasing from 30 to all 102 FLEURS languages and testing on conversational or code-switched speech.
- Tokenizer redesign for SALMONN-class models: retraining the embedding layer with better coverage of low-resource scripts (byte-level BPE with explicit reserved tokens for Khmer, Armenian, etc.).

6. Conclusion

This project investigated how three contemporary multimodal foundation models — SeamlessM4T, SONAR, and SALMONN — internally encode the same semantic content across 30 languages and across the speech and text modalities. Using Singular Vector Canonical Correlation Analysis (SVCCA) on activations from the FLEURS test set, we measured three complementary similarity scores per layer: intra-lingual cross-modal, cross-lingual text, and cross-lingual speech.

The analysis yielded four headline findings:

- Cross-modal representations converge over layers, except in the very first layers where modality-specific surface features dominate.
- Length adaptation is crucial for reducing the cross-modal gap, but its effectiveness is mostly limited to high- and medium-resource languages; only SONAR's explicitly aligned pooling helps low-resource languages meaningfully.
- Speech exhibits larger cross-lingual differences than text, owing to its intrinsically greater surface variability (accent, speaker, pace, recording conditions).
- For models not explicitly trained for modality-agnostic representations, the modality gap is more prominent than the language gap. SONAR's explicit alignment training inverts this ordering.

Beyond these scientific findings, the project produced two practical contributions:

- A reproducible Colab-compatible pipeline that loads each model on a single T4 GPU, extracts per-layer activations, and computes SVCCA scores in under a few GPU-hours per model.
- Concrete recommendations for practitioners: when building speech-text systems for low-resource or zero-shot use cases, initialize from foundation models explicitly trained for modality- and language-alignment (such as SONAR), rather than from larger but unaligned general-purpose multimodal LLMs.

Finally, this project demonstrates that intrinsic representation analysis is a cheap, model-agnostic diagnostic that can surface important issues — tokenizer fragmentation, weak low-resource alignment, residual modality gaps — well before they manifest as downstream-task failures. Such analyses should be incorporated into the standard development cycle of any new multimodal foundation model.

7. References

- [1] H. Lee, D. Liu, S. Sinhamahapatra, and J. Niehues, "How do multimodal foundation models encode text and speech?" NAACL-HLT 2025, pp. 600-610.
- [2] Seamless Communication et al., "SeamlessM4T - massively multilingual & multimodal machine translation," arXiv:2308.11596, 2023.
- [3] P.-A. Duquenne, H. Schwenk, and B. Sagot, "SONAR: sentence-level multimodal and language-agnostic representations," arXiv:2308.11466, 2023.
- [4] C. Tang et al., "SALMONN: towards generic hearing abilities for large language models," ICLR 2024.
- [5] M. Raghu, J. Gilmer, J. Yosinski, and J. Sohl-Dickstein, "SVCCA: singular vector canonical correlation analysis for deep learning dynamics," NeurIPS 2017, pp. 6076-6085.
- [6] A. Conneau et al., "FLEURS: few-shot learning evaluation of universal representations of speech," IEEE SLT Workshop 2023, pp. 798-805.
- [7] N. Goyal et al., "The FLORES-101 evaluation benchmark," TACL, vol. 10, pp. 522-538, 2022.
- [8] NLLB Team et al., "No language left behind: scaling human-centered machine translation," arXiv:2207.04672, 2022.
- [9] Y. Belinkov and J. R. Glass, "Analyzing hidden representations in end-to-end ASR systems," NeurIPS 2017, pp. 2441-2451.
- [10] A. Conneau et al., "Unsupervised cross-lingual representation learning at scale," ACL 2020, pp. 8440-8451.
- [11] Z. Wu et al., "The semantic hub hypothesis: language models share semantic representations across languages and modalities," arXiv:2411.04986, 2024.
- [12] C. Wendler et al., "Do llamas work in English? On the latent language of multilingual transformers," ACL 2024, pp. 15366-15394.
- [13] Y. Chu et al., "Qwen2-Audio technical report," arXiv:2407.10759, 2024.
- [14] A. Dubey et al., "The Llama 3 herd of models," arXiv:2407.21783, 2024.
- [15] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," JMLR, vol. 9, pp. 2579-2605, 2008.

8. Appendix: Project Code

The complete Google Colab notebook used for all experiments in this project is available at:

<https://colab.research.google.com/drive/1xXpVw-fpl36CcCfE9G19c8XnDerElWij?usp=sharing>

The notebook includes:

- Full implementation of SVCCA similarity measurement
- Preprocessing pipeline for FLEURS dataset (30 languages)
- Activation extraction for SeamlessM4T, SONAR, and SALMONN
- All visualizations (line plots, bar charts, t-SNE projections)
- Reproducible results with fixed random seed